Fiscal Year:	FY 2005	**Task Last Updated:**	FY 12/20/2004
PI Name:	German, Randall M. Ph.D.		
Project Title:	Gravitational Effects on Distortion in Sintering		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Metals and alloys		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	randgerman@gmail.com	**Fax:**	FY none
PI Organization Type:	UNIVERSITY	**Phone:**	858-922-4985
Organization Name:	San Diego State University		
PI Address 1:	College of Engineering		
PI Address 2:	282 Surfview Court		
PI Web Page:			
City:	Del Mar	**State:**	CA
Zip Code:	92014	**Congressional District:**	53
Comments:	co-investigator Eugene A Olevsky, SDSU, Elisa Torresani, SDSU, MS 1326		
Project Type:	Flight	**Solicitation / Funding Source:**	96-HEDS-02
Start Date:	02/01/1998	**End Date:**	12/31/2006
No. of Post Docs:	3	**No. of PhD Degrees:**	5
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Palosz, Witold	**Contact Phone:**	256 544-1272
Contact Email:	Witold.Palosz@msfc.nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: original end date was 12/31/2010 (jp 2/08)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG8-1452		
Performance Goal No.:	1.25		
Performance Goal Text:	Use microgravity to establish and improve quantitative and predictive relationships between the structure, processing, and properties of materials.		

Task Description:	Particle-based fabrication studies using a viscous or liquid phase for particle bonding with low-gravity conditions that impact densification and distortion. These studies are performed using systematic variations of the controlling parameters of porosity, solid:liquid ratio, and basic thermodynamic features such as solubility and dihedral angle. Further, to examine the role of porosity and pore size (and associated buoyancy effcts) on sintering densification and distortion, with attention to formation of large pore agglomerates, possible distortion due to pore buoyancy, relations between distortion and densification, and assessment of capillary effects on both. Calculation and measurement of in situ strength and viscosity. Use of the evolving mechanical property data to feed into finite element and fluid dynamic calculations to predict component size, shape, and defects during sintering densification. To support this model, quenching experiments are being used to freeze the evolving microstructures to measure density and pore saturation for identification of possible options to resist distortion during sinteirng. Extraction of gravity effect is important since there three forces causing distortion during densification - anisotropic gravity forces, and hydrostatic forces from capillarity, and density (porosity and pore size) gradients. One of the findings is that we can now manipulate the sintering cycle to improve strength and avoid distortion, a finding that relates to simple changes in thermal cycle design and should therefore have wide industrial appeal. 1) Investigate the effect of porosity on the microstructure and compact shape retention during sintering in the presence of a liquid phase, with separation of gravitational role via pore buoyancy. 2) Systematically and independently vary the pore size and porosity level and location to assess pore effects on distortion or slumping during liquid phase sintering while examining a range of dihedral angles, solid:liquid ratios, and sintering time effects. 3) Investigate buoyancy-induced pore migration as part of the densification process to determine if vacancy annhilation is the only densification process; further evaluate pore migration effects on induced distortion. 4) Time resolved determination of the sequence of events and interaction of events leading to distortion and densification, thereby extracting any possible separation of the two events for practical benefits. 5) Determine if there are unique thermal strategies to avoid distortion during sintering, especially during the periods of low strength as isolated in the experiments and modeling. 6) Demonstration of scaling laws to model proposed systems in various gravitational environments to improve dimensional precision and to predict final size, shape, and properties. Fundamentally, sintering is an important net-shaping technology applied to metals, ceramics, and cermets, composites, cemented carbides and other mixed powder systems. Sintering approaches are a favorite to freeform fabrication routes, many of which are being considerd for in-space fabrication and repair. Further, recent efforts have suggested sintering as a means to use space-based materials for fabrication of living, launch, and other structures. In the USA, sintered materials contribute over $23 billion to the economy. This research is applicable to improved sintering cycles and the production of higher quality, more precise components. It combines intentional green body changes in composition, porosity, pore size, and liquid content with the presence or absence of gravity to develop our understanding in this technologically significant area. The source of component distortion in sintering and the time-dependent role of porosity are poorly understood in sintering practice. Pores are very stable under microgravity conditions. Consequently, microgravity provides the unique opportunity to control and manipulate pores to extract the role of porosity and pore size with respect to densification and distortion. Further, changes in coposition and thermal cycle are being used to improve the component rigidity (strength) at temperature via both modeling and experimental efforts. We have determined that nondestructive tests can now possibly provide needed information for thermal cycle design to minimize distortion.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study has implications in all aspects of human activity on earth. For example, polar ice caps contain trapped atmospheres from past times, providing a means to determin the greenhouse changes if samples can be recovered and dated. Sintering models used in the current project have been trained for ice (snow) sintering to provide a calculation as to the time when pores closed versus depth (gravitational head) and annual snowfall. Likewise, the models developed in this work show how mankind can fabricate precise components, thereby minimizing wasted materials and efforts in the fabrication of industrial compnents. With respect to the burden of space launch, this work provides fundamental data on the role of gravity (and microgravity) with respect to various proposed in-space fabrication and repair routes. The cost of launching various freeform fabrication devices is enormous, both in direct cost and in space operating difficulties. This work is developing accurate models for final size and shape that can be mapped onto the operating parameters of various proposed freeform fabrication and repair ideas to show realistic feasiblity, without the burden of launch.
Task Progress:	The GEDS team is approaching completion of all initial tasks and is now working aggressively on engineering issues associated with simplification of the experimental protocol. Over the past year, the number of unique experiments has been decreased as refined ground-based data has become available. Further, progress has been made in advising various in-space fabrication and repair groups on the feasiblity of their sintering concepts. This has further been extended to collaboration in polar ice sintering models and lunar regolith sintering models. Most recently the team has been focused on intense review of safety issues associated with the planned experiments. This has first resulted in a major reduction in hazard or toxicity concerns. Such a reduction came about from laboratory measures of outgassing, vapor egression, and weight changes with various sample and crucible prepartion efforts. The resulting effort has quantified sources of evaporation constituents and means to remove such contaminants prior to flight. Compuer modeling reach full three-dimensional realization for liquid phase sintering and critical publications were generated. The time-resolved transient events associated with liquid formation in the tungsten heavy alloys was captured and quantified. That work was submitted for journal publication. It shows the conditions under which the structure weakens and potentially deforms on liquid formation. Accordingly, new heating strategies emerge from this insight. Vacuum sealing experiments have been evaluated and the crucible design has been slightly modified to accommodate the need for low vapor pressures during sintering. Many of these requirements were relatively new to the program, reflecting internal changes at NASA. However, the GEDS team was successful in addressing these challenges. To assist in defiining the mechanical properties possible from sintering in space (in-space fabrication and repair), the

	GEDS team has been designing experiments to gather the first ever mechanical properties of liquid phase sintered alloys fabricated both under ground and microgravity conditions. This is critically needed data, since to date we have indications that in-space fabricated material will be weaker and will require special treatments to match the strength, ductility, and toughness attributes of materials they are replacing. Finally, the complexity and number of experiments are constantly being reviewed with the intent of reducing the experimental cost. A dramatic decrease in sintering runs has taken place. Further, experiments were initiated this past year to lower the peak temperature to gather the same science gains with less danger to furnace thermocouples. This latter effort was not completed, but new tungsten-based alloys are under testing. I successful, this will remove a major concern on furnace lifetime in space.
Bibliography Type:	Description: (Last Updated: 03/27/2025)
Abstracts for Journals and Proceedings	Ozkal, B.; Upadhyaya, A; Ovecoglu, M. L.; German, R. M. "Realtime Sintering Observations in W-Cu System: Accelerated Rearrangement Densification via Copper Coated Tungsten Powders Approach" Proceedings PM 2004, European Powder Metallurgy Association, Shrewsbury, UK. published on compact disk, no page numbers. , Oct-2004
Articles in Peer-reviewed Journals	Bollina R.; German R. M. "Heating rate effects on microstructural properties of liquid phase sintered tungsten heavy alloys" ” International Journal of Refractory Metals and Hard Materials, 22: 117-127, 2004. , Jun-2004
Articles in Peer-reviewed Journals	Liu, J.; German, R. M. "Three-Dimensional Coordination Number in Liquid Phase Sintered Microstructures from Two-Dimensional Connectivity" P/M Science and Technology Briefs, 6 [2]: 19-21, 2004. , Jun-2004
Articles in Peer-reviewed Journals	Binet, C.; Lencoski, K. L.; Heaney, D. F.; German, R. M. "Modeling of Distortion after Densification during Liquid Phase Sintering" Metallurgical and Materials Transactions, 35A: 3833-3841, 2004. , Nov-2004
Awards	German, R. M. "Protoclo del Solemne Acto del dia de la Universidad del Curso 2003/2004, Investidura Como Doctor Honoris Causa de Prof. Dr. D. Randall German, Universidad Carlos III de Madrid, Getafe, Madrid, Spain, 20 February 2004." Feb-2004
Papers from Meeting Proceedings	German, R. M.; Suri, P.; Erhardt, N.; Chung, S. H. " “Gravitational Effects on Distortion in Liquid Phase Sintering" Advances in Powder Metallurgy and Particulate Materials - 2004, part 5, Metal Powder Industries Federation, Princeton, NJ, 41-50, 2004. Nov-2004
Papers from Meeting Proceedings	German, R. M.; Blaine, D.; Chung, S. H. "“Distortion and Densification Control During Liquid Phase Sintering of High Performance Materials” " invited presentation, Eight International Conference on Numerical Methods in Industrial Forming Processes, Ohio State University, Columbus, Ohio, 14 June 2004. Jun-2004
Papers from Meeting Proceedings	German, R. M.; Chung, S. H.; Erhardt, N.; Suri, P. "“Gravitational Effects on Distortion in Liquid Phase Sintering”" presented at the Powder Metallurgy and Particulate Materials Conference PM2Tech, Metal Powder Industries Federation, Chicago, Illinois, 16 June 2004 (with N. S. H. Chung, and Erhardt, P. Suri). Jun-2004
Papers from Meeting Proceedings	German, R. M.; Blaine, D.; Suri, P.; Chung, S. H.; Park, J. S. "“Integral Work of Sintering Concepts Applied to Liquid Phase Sintering Densification, Distortion, and Microstructure Evolution” " invited presentation PM 2004 Powder Metallurgy World Congress, European Powder Metallurgy Association, Vienna, Austria, 19 October 2004 (with D. Blaine, P. Suri, S. H. Chung, S. J. Park). Oct-2004
Presentation	German, R. M. "“Current Techniques in Powder Metallurgy: Liquid Phase Sintering in DOD Applications” " invited presentation, Air Force Research Laboratory, Edwards Air Force Base, California, 26 January 2004. Jan-2004
Presentation	German, R. M. "“Modeling Densification and Distortion in Liquid Phase Sintering” " invited seminar, Rockwell Scientific, Thousand Oaks, California, 17 February 2004. Feb-2004
Presentation	German, R. M. "“Computer Analysis and Engineering - Sintering Models”" invited presentation, Extrude Hone Corp., Irwin, Pennsylvania, 19 July 2004. Jul-2004
Presentation	German, R. M. "“Grain Boundary Wetting Transients in Liquid Phase Sintering”" invited presentation, Gordon Research Conference on Solid State Studies in Ceramics, Kimball Union Academy, Meriden, New Hampshire, 9 August 2004. Aug-2004
Presentation	German, R. M. "“Thermal Management and Packaging Design Using Powder Injection Molding” " invited seminar, Sun Microsystems, San Diego, California, 11 March 2004. Mar-2004
Presentation	Chung, S. H; German, R. M.; Bollina, R.; Suri, P.; Kwon, Y. S. "“Study on the Effects of Liquid Volume Fraction on Sintering Behavior of W-Ni-Fe Alloy” " presented at PIM 2004 International Conference on Powder Injection Molding of Metals, Ceramics, and Carbides, Orlando, Florida, 23 March 2004. Mar-2004
Presentation	German, R. M. "“Liquid Phase Sintering - From Basics to Applications”" invited seminar, Chemical and Materials Engineering, University of California, Davis, California, 8 April 2004. Apr-2004

Presentation	German, R. M. "“Distortion and Densification Control during Sintering” " invited seminar, Metallurgy and Materials Engineering Department, University of Nevada, Reno, Nevada, 9 April 2004. Apr-2004
Presentation	German, R. M. "“Net Shape Fabrication Using Advanced Powder Technologies” " invited seminar, Mechanical Engineering Department, San Diego State University, San Diego, California, 15 April 2004. Apr-2004
Presentation	German, R. M. "“Net Shape Fabrication Using Advanced Powder Technologies”" invited seminar, Materials Science and Engineering Department, University of California, Los Angeles, California, 16 April 2004. Apr-2004
Presentation	German, R. M. "“Emerging Powder Metallurgy and Particulate Material Processes”" invited graduate seminar, Mechanical Engineering Department, Tsinghua University, Beijing, China, 19 April 2004. Apr-2004
Presentation	German, R. M. "“Emerging Powder Metallurgy Processes” " invited seminar, Materials Science and Engineering Department and State Key Laboratory for Powder Metallurgy, Central South University, Chengsha, Hunan, China, 24 April 2004. Apr-2004
Presentation	German, R. M. "“Computer Analysis and Engineering- Models” " invited technology seminar, H. C. Starck, Newton, Massachusetts, 25 May 2004. May-2004